

KALINGA INSTITUTE OF INDUSTRIAL TECHNOLOGY

Deemed to be University, Bhubaneswar - 751024, Odisha

MID SEMESTER EXAMINATION — SOLUTIONS & MARKING SCHEME

Subject: Data Structure

Time: 1.5 Hours

Semester / Programme: 3rd Semester

Full Marks: 20

Instructions: Answer Question No. 1 (Compulsory) and any THREE from Question No. 2 to 5. Figures in brackets indicate marks.

SECTION - A (Compulsory)

1(a) How many different Binary Search Trees (BST) can be constructed with 4 distinct keys? [1]

$$C_n = \frac{1}{n+1} \binom{2n}{n}$$

Solution & Key Concept:

- Use the Catalan number formula: $C_n = (1 / (n+1)) C_{2n}^n$
- Substitute $n=4$: $C_4 = (1 / 5) C_8^4 = 14$

Answer: 14

Marking Rubric: [1 Mark] for correct formula and final answer

1(b) Write the postfix expression for the following infix expression: $A + B \times C - D / E$ [1]

Solution & Key Concept:

- Multiplication and division first: $B \times C \rightarrow BC*$
- Full postfix: $ABC*+DE/-$

Answer: $ABC*+DE/-$

Marking Rubric: [1 Mark] for correct conversion

1(d) State the worst-case time complexity of searching an element in a Binary Search Tree (BST) of height h . [1]

$$T(n) = O(h)$$

Solution & Key Concept:

- Traversing from root to deepest leaf in skewed tree

Answer: $O(h)$

Marking Rubric: [1 Mark] for correct bound

1(e) What is the maximum number of nodes in a binary tree of height h (assuming root is at height 0)? [1]

$$N_{max} = 2^{h+1} - 1$$

Solution & Key Concept:

- Sum of geometric progression $\sum_{i=0}^h 2^i = 2^{h+1} - 1$

Answer: $2^{h+1} - 1$

Marking Rubric: [1 Mark] for correct formula

SECTION - B (Answer any THREE)

Answer any THREE questions from Q.2 to Q.5. Each question carries 5 marks.

Q.2. (a) Explain array storage representation in row-major order. [2.5]

$$Loc(A[i][j]) = B + (i \times N + j) \times w$$

Solution for (a):

- Base address plus 2D offset

Result / Conclusion: Address formula verified

Marking Rubric: [1.5 Marks] formula derivation, [1 Mark] definition

(b) Evaluate the postfix expression: 6\ 2\ 3\ +\ -\ 3\ 8\ 2\ /\ +\ *\ 2\ /\ 3\ +. [2.5]

Solution for (b):

- Stack step-by-step evaluation

Result / Conclusion: Computed numerical result

Marking Rubric: [2.5 Marks] complete stack tracing

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